PROJECT PHOENIX

REINFORCED CONCRETE BEAM PRELIMINARY DESIGN REPORT

Beam: RCB-001 - Simply supported demonstration beam
Project: PHX-RC-DEMO-001
Status: PRELIMINARY STRUCTURAL DESIGN - ENGINEER REVIEW REQUIRED

Geometry: L=5.0 m, b=300 mm, h=500 mm
Materials: C30/37 / B500B
Standard profile: NL_EC2_2005_A1_2015_NA_A2_2025_PRELIMINARY

This report is not a signed structural calculation and is not approved for construction.

1. LOADS AND ANALYSIS

Self weight: 3.75 kN/m
Total permanent UDL: 11.75 kN/m
Variable UDL: 10.0 kN/m
ULS reaction A: 92.156 kN
ULS reaction B: 92.156 kN
ULS maximum moment: 133.945 kNm
ULS maximum shear: 92.156 kN

Point loads:
- P-01: 20.0 kN at 2.5 m

2. FLEXURE AND SHEAR

Required steel: 737.814 mm2
Provided steel: 804.248 mm2
Bottom reinforcement: 4T16 bottom (As=804 mm2; clear=50 mm)
Moment resistance: 146.006 kNm
Flexure utilization: 0.917

Concrete shear resistance: 70.519 kN
Stirrups: 2-leg T8 stirrups @ 175 mm
Governing shear resistance: 100.931 kN
Shear utilization: 0.913

3. SERVICEABILITY AND DETAILING

Estimated deflection: 8.311 mm
Deflection limit: 20.0 mm
Estimated crack width: 0.134 mm
Crack width limit: 0.3 mm
Design anchorage length: 532 mm
Available anchorage length: 850 mm

The serviceability calculations are preliminary estimates and require engineer confirmation.

4. DESIGN CHECKS

CHK-001 | Statics | PASS | Reactions equal total ULS load.
CHK-002 | Flexure resistance | PASS | MEd/MRd = 0.917
CHK-003 | Neutral axis | PASS | x/(xlim*d) = 0.389
CHK-004 | Minimum reinforcement | PASS | As,prov = 804.2 mm2; As,min = 203.1 mm2
CHK-005 | Maximum reinforcement | PASS | As,prov = 804.2 mm2; As,max = 6000.0 mm2
CHK-006 | Bar fit | PASS | 4T16 bottom (As=804 mm2; clear=50 mm)
CHK-007 | Shear resistance | PASS | VEd/VRd = 0.913
CHK-008 | Shear compression strut | PASS | VEd = 92.2 kN; VRd,max = 544.1 kN
CHK-009 | Deflection | PASS | delta = 8.31 mm; limit = 20.00 mm
CHK-010 | Crack width | PASS | wk = 0.134 mm; limit = 0.300 mm
CHK-011 | Anchorage fit | PASS | lb,design = 532 mm; available = 850 mm
CHK-012 | Support bearing | PASS | bearing utilization = 0.100
CHK-013 | Deep-section review gate | PASS | h = 500 mm; mandatory review threshold = 700 mm
CHK-014 | Fire design boundary | PASS | Fire design is excluded and must be completed separately.
CHK-015 | Professional review gate | PASS | All outputs remain preliminary until signed by a competent structural engineer.
CHK-016 | Standard profile declared | PASS | NL_EC2_2005_A1_2015_NA_A2_2025_PRELIMINARY

5. LIMITATIONS AND NEXT GATE

- This is a preliminary design engine and not a signed structural calculation.
- The selected standard profile and National Annex parameters must be confirmed for the project.
- Fire resistance, robustness, seismic design, fatigue and accidental actions are outside this v1.0.0 scope.
- Construction tolerances, bar curtailment, laps, couplers, support-zone detailing and actual load paths require engineer review.
- The 2025 Dutch National Annex introduced shear-related changes; deep sections trigger a mandatory review gate.

Next gate:
Assign the beam to an approved model object, replace example loads with project loads, and obtain signed structural-engineer review before use for permit, tender or execution.